Probability basics

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Defining probabilities

Definition 0.1 (Random experiment)**.** A **random experiment** is a procedure that produces exactly one outcome out of several possible outcomes, where which outcome it will produce is not determined in advance.

Remark. The possible outcomes are collected in the sample space. Physical procedures, such as rolling a die, and sampling procedures, such as picking a unit from a data set without looking, are both random experiments.

Example 0.1 (Picking a car model from the Auto data)**.** The Auto data record the number of engine cylinders and the fuel economy (miles per gallon) of 392 car models sold between 1970 and 1982. They come from the StatLib library at Carnegie Mellon University and ship with the book *An Introduction to Statistical Learning* and the R package ISLR2 (`ISLR2::Auto`). This site has a copy at `data/auto.csv`.
Pick one car model from the data without looking, and record its number of cylinders. Which model comes up is not determined in advance, so this procedure is a random experiment. Its possible outcomes are the cylinder counts that occur in the data:

```
# read the local copy of the Auto data (one row per car model):
auto <- read.csv("data/auto.csv")

# the possible outcomes are the distinct cylinder counts in the data:
sort(unique(auto$cylinders))
#> [1] 3 4 5 6 8
```

Definition 0.2 (Sample space)**.** The **sample space** of a random experiment, denoted Ω, is the set of all of its possible outcomes.

Example 0.2 (Sample space of a die roll)**.** Rolling a six-sided die once has sample space $\Omega = \{1, 2, 3, 4, 5, 6\}$.

Definition 0.3 (Event)**.** An **event** is a subset of the sample space Ω: a set of outcomes.

Remark. When Ω is uncountable, such as an interval of real numbers, only the subsets in a designated σ-algebra $\mathcal{S}$ on Ω count as events (some sources call a σ-algebra a **Borel field** (Casella and Berger 2002, Def. 1.2.1)); every set that arises in these notes is in $\mathcal{S}$. When Ω is finite or countably infinite, every subset of Ω can be an event, because the collection of all subsets of Ω is a σ-algebra (all subsets form a σ-algebra), as in σ-algebras for die rolls.

Example 0.3 (Rolling an even number)**.** In Example 0.2, the event "the roll is even" is $A = \{2, 4, 6\}$.

Definition 0.4 (Occurrence of an event)**.** An event A **occurs** when the outcome of the experiment is in A, and does not occur when the outcome is not in A.

Example 0.4 (Rolling a 4)**.** In Example 0.3, if the die shows 4, the event "the roll is even", $\{2, 4, 6\}$, occurs, because $4 \in \{2, 4, 6\}$, and the event "the roll is at most 2", $\{1, 2\}$, does not occur.

Definition 0.5 (Complement of an event)**.** The **complement** of an event A, denoted $\neg A$, is the event that A does not occur: the set of outcomes in the sample space Ω that are not in A.

$$\neg A \stackrel{\text{def}}{=} \Omega \setminus A$$

Remark. Other sources write A^c or $\bar{A}$.

Example 0.5 (Rolling an odd number)**.** In Example 0.3, the complement of the event "the roll is even", $A = \{2, 4, 6\}$, is the event "the roll is odd", $\neg A = \{1, 3, 5\}$.

Definition 0.6 (Mutually exclusive events)**.** Finitely or countably many sets $A_1, A_2, ...$, such as events, are **mutually exclusive** (also called **disjoint**, **pairwise disjoint**, or **mutually disjoint**) when no two of them share an element:

$$A_i \cap A_j = \emptyset \quad \text{for all } i \neq j$$

Remark. The math notes define the same property for sets in general, as pairwise disjoint.

Example 0.6 (Low and high rolls)**.** In Example 0.2, the events "the roll is at most 2", $\{1, 2\}$, and "the roll is at least 5", $\{5, 6\}$, are mutually exclusive: no roll is in both. The events "the roll is even", $\{2, 4, 6\}$, and "the roll is at most 2", $\{1, 2\}$, are not mutually exclusive: a roll of 2 is in both.

Definition 0.7 (Partition of an event)**.** A **partition** of an event A is a finite or countably infinite collection of events $A_1, A_2, \ldots$ that satisfies:

- The events $A_1, A_2, \ldots$ are mutually exclusive.
- Their union is A: $\bigcup_i A_i = A$.

A partition of the sample space is the case $A = \Omega$.

Remark. Some sources also require each piece A_i to contain at least one outcome.

Example 0.7 (Partitioning the die roll)**.** In Example 0.2, the events "the roll is even", $\{2, 4, 6\}$, and "the roll is odd", $\{1, 3, 5\}$, are mutually exclusive, and their union is Ω, so they partition the sample space. So do the six single-outcome events $\{1\}, \{2\}, \ldots, \{6\}$. The events $\{1, 2\}$ and $\{5, 6\}$ from Example 0.6 do not partition Ω: their union leaves out 3 and 4.

Definition 0.8 (Probability measure)**.** A **probability measure** on a sample space Ω, often denoted $\Pr()$ or $\mathrm{P}()$, is a measure on the events of Ω that gives the whole sample space probability 1:

$$\Pr(\Omega) = 1$$

Remark. Other sources call a probability measure a **probability function** (Casella and Berger 2002, Def. 1.2.4). The two names mean the same thing.

Example 0.8 (Probability measure for a fair die)**.** For the die roll in Example 0.2, define $\Pr(A) \stackrel{\text{def}}{=} |A|/6$, where $|A|$ is the number of outcomes in A. This function is $1/6$ times the

counting measure $\mu(A) = |A|$ on the die rolls (counting elements is a measure), so it is a measure on the events of Ω, and it gives

$$\begin{aligned}\Pr(\Omega) &= 6/6 \\ &= 1.\end{aligned}$$

The event "the roll is even" from Example 0.3 has probability

$$\begin{aligned}\Pr(\{2,4,6\}) &= 3/6 \\ &= 1/2.\end{aligned}$$

Corollary 0.1 (Probability measures are finitely additive)**.** *Every probability measure is finitely additive: for any mutually exclusive events* $A_1, \ldots, A_n$*,*

$$\Pr(A_1 \cup \cdots \cup A_n) = \sum_{i=1}^{n} \Pr(A_i)$$

Proof

Proof. A probability measure is a measure (Definition 0.8), so measures are finitely additive applies to it. □

Theorem 0.1 (Continuity of probability)**.** *If* $A_1 \supseteq A_2 \supseteq \cdots$ *are events with* $\bigcap_{i=1}^{\infty} A_i = \emptyset$*, then:*

$$\lim_{n\to\infty} \Pr(A_n) = 0$$

Proof

Proof. For each i, let $B_i \stackrel{\text{def}}{=} A_i \setminus A_{i+1}$, the outcomes in A_i but not in A_{i+1}. Each $B_i = A_i \cap (\Omega \setminus A_{i+1})$ is an event, by closure properties of a σ-algebra.
The B_i *are pairwise disjoint.* For $i < j$, $B_j \subseteq A_j \subseteq A_{i+1}$, while B_i has no outcomes in A_{i+1}, so $B_i \cap B_j = \emptyset$.
$A_n = \bigcup_{i=n}^{\infty} B_i$ *for each* n. For $i \geq n$, $B_i \subseteq A_i \subseteq A_n$, so the union is contained in A_n. Conversely, let $\omega \in A_n$. Since $\bigcap_i A_i = \emptyset$, ω is not in every A_i; since the A_i are nested, the indices i with $\omega \in A_i$ are $1, \ldots, m$ for some $m \geq n$. Then $\omega \in A_m$ and $\omega \notin A_{m+1}$, so $\omega \in B_m$.
$\Pr(A_1)$ *is finite.* The events A_1 and $\Omega \setminus A_1$ are disjoint with union Ω, so Corollary 0.1 applies to them:

$$\begin{aligned}
\Pr(A_1) &\le \Pr(A_1) + \Pr(\Omega \setminus A_1) && (\Pr(\Omega \setminus A_1) \ge 0) \\
&= \Pr(\Omega) && (\text{finite additivity of } \Pr) \\
&= 1 && (\text{definition of a probability measure})
\end{aligned}$$

The limit. For each n:

$$\begin{aligned}
\Pr(A_n) &= \Pr\left(\bigcup_{i=n}^{\infty} B_i\right) && \left(A_n = \bigcup_{i=n}^{\infty} B_i\right) \\
&= \sum_{i=n}^{\infty} \Pr(B_i) && (\text{countable additivity; the } B_i \text{ are pairwise disjoint})
\end{aligned}$$

With $n = 1$, this says the series $\sum_{i=1}^{\infty} \Pr(B_i)$ has the finite total $\Pr(A_1)$. So, for $n \ge 2$:

$$\begin{aligned}
\Pr(A_n) &= \sum_{i=1}^{\infty} \Pr(B_i) - \sum_{i=1}^{n-1} \Pr(B_i) && (\text{split off the first } n-1 \text{ terms; the total is finite}) \\
&= \Pr(A_1) - \sum_{i=1}^{n-1} \Pr(B_i) && (\text{the case } n = 1)
\end{aligned}$$

As $n \to \infty$, the partial sums $\sum_{i=1}^{n-1} \Pr(B_i)$ converge to the total $\Pr(A_1)$, so $\Pr(A_n) \to \Pr(A_1) - \Pr(A_1) = 0$. □

Remark. Requiring countable additivity, not just finite additivity, enables results such as Theorem 0.1, and it is what makes sums over countably infinite partitions valid.

Theorem 0.2 (Kolmogorov axioms)**.** *A function* Pr *that assigns a real number* $\Pr(A)$ *to each event* A *of a sample space* Ω *is a probability measure if and only if it satisfies:*

1. *For any event* A, $\Pr(A) \ge 0$.
2. *The probability of the whole sample space is 1:*

$$\Pr(\Omega) = 1$$

3. Pr *is countably additive: for any mutually exclusive events* $A_1, A_2, \ldots,$

$$\Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \Pr(A_i)$$

Proof

Proof. Suppose Pr is a probability measure. It is a measure, so its values lie in the extended non-negative reals $[0, \infty]$, which gives axiom 1, and it is countably additive, which is axiom 3. Axiom 2 is the condition $\Pr(\Omega) = 1$ in Definition 0.8.
Conversely, suppose Pr satisfies the three axioms. By axiom 1, Pr takes values in $[0, \infty]$, so axiom 3 says Pr is countably additive in the sense of the definition of countable additivity. Then, by countable additivity and the empty set, $\Pr(\emptyset)$ is 0 or ∞; $\Pr(\emptyset)$ is a real number, so $\Pr(\emptyset) = 0$. So Pr satisfies both conditions of the definition of a measure ($\Pr(\emptyset) = 0$ and countable additivity) and is a measure on the events of Ω. With axiom 2, Pr is a probability measure (Definition 0.8). □

Remark. Many sources define a probability measure by these three axioms instead; the theorem shows that the two definitions agree. The axioms are named for Kolmogorov, who introduced them in 1933 (see Wikipedia: Probability axioms). The axiom form does not list $\Pr(\emptyset) = 0$; the proof above derives it.

Theorem 0.3 (Probability of a subset's intersection)**.** *If A and B are events and $A \subseteq B$, then $\Pr(A \cap B) = \Pr(A)$.*

Proof

Proof. Since $A \subseteq B$, every outcome in A is also in B, so $A \cap B = A$, and therefore $\Pr(A \cap B) = \Pr(A)$. □

Theorem 0.4 (An event and its complement sum to 1)**.** *For any event A and its complement $\neg A$:*

$$\Pr(A) + \Pr(\neg A) = 1$$

Proof

Proof. The events A and $\neg A$ are disjoint, and their union is Ω.

$$\begin{aligned} \Pr(A) + \Pr(\neg A) &= \Pr(A \cup \neg A) && \text{(additivity of probability for disjoint events)} \\ &= \Pr(\Omega) && (A \cup \neg A = \Omega) \\ &= 1 && \text{(probability of the sample space is 1)} \end{aligned}$$

□

Corollary 0.2 (Complement rule)**.** *For any event A:*

$$\Pr(\neg A) = 1 - \Pr(A)$$

Proof

Proof. Subtract $\Pr(A)$ from both sides of Theorem 0.4. □

Corollary 0.3 (Complement rule in probability (π) notation)**.** *If the probability of an event A is $\Pr(A) = \pi$, then the probability that A does not occur is:*

$$\Pr(\neg A) = 1 - \pi$$

Proof

Proof.

$$\begin{aligned} \Pr(\neg A) &= 1 - \Pr(A) && \text{(complement rule)} \\ &= 1 - \pi && \text{(substitute } \Pr(A) = \pi\text{)} \end{aligned}$$

□

Example 0.9 (Probability of not rolling a six)**.** For a fair die, the event "the roll is a six" has probability $\pi = 1/6$, so by Corollary 0.3 the probability of not rolling a six is $1 - 1/6 = 5/6$.

Definition 0.9 (Choosing at random)**.** Let S be a set with a measure μ such that $0 < \mu(S) < \infty$. To **choose an element of S at random** is to run a random experiment whose sample space is S and whose probability measure is the uniform distribution on S.

Example 0.10 (Choosing at random from a finite set and from an interval)**.** For a finite set S, let $\mu(A) = |A|$ be the counting measure. Then $\Pr(A) = |A|/|S|$, so every element of S has the same probability $1/|S|$.
For the interval $S = [0, 1]$, let $\mu(A)$ be the length of A. Then $\Pr(A)$ is the length of A, so the probability that the chosen number lies in $[0.2, 0.5]$ is 0.3. The chosen number has the Uniform$(0, 1)$ distribution.
The integers have infinite counting measure, so they have no choice at random in this sense.

Example 0.11 (Choosing a car at random from the Auto data)**.** The Auto data of Example 0.1 have 392 car models. Let S be the set of car models in the data. Choosing a car at random from S (Definition 0.9), with the counting measure, gives each car probability $1/|S|$, so the probability that the car has four cylinders is the fraction of cars with four cylinders:

```
# read the local copy of the Auto data (one row per car model):
auto <- read.csv("data/auto.csv")

# the event "four cylinders" is the set of rows where cylinders == 4;
# the mean of the logical vector is |A| / |S|, the probability in
# the definition of choosing at random:
mean(auto$cylinders == 4)
#> [1] 0.5076531
```

Conditional probability

Definition 0.10 (Conditional probability)**.** For two events A and B with $\Pr(B) > 0$, the **conditional probability** of A given B, denoted $\Pr(A \mid B)$, is:

$$\Pr(A \mid B) \stackrel{\text{def}}{=} \frac{\Pr(A \cap B)}{\Pr(B)}$$

Remark. Geometrically and computationally, conditioning on an event B can be viewed as *slicing* the sample space to retain only the outcomes compatible with B, and then *renormalizing* the remaining probabilities (dividing by $\Pr(B)$) so that they sum to 1 (Ross 2022, sec. 2.7.4).

Example 0.12 (Rolling a six, given an even roll)**.** For a fair die (Example 0.8), let $A = \{6\}$ and $B = \{2, 4, 6\}$. Then $A \cap B = \{6\}$, so:

$$\begin{aligned} \Pr(A \mid B) &= \frac{\Pr(A \cap B)}{\Pr(B)} && \text{(definition of conditional probability)} \\ &= \frac{1/6}{3/6} && \text{(substitute the probabilities)} \\ &= \frac{1}{3} && \text{(simplify)} \end{aligned}$$

Theorem 0.5 (Law of conditional probability)**.** *For any two events A and B with $\Pr(B) > 0$:*

$$\Pr(A \cap B) = \Pr(A \mid B) \cdot \Pr(B)$$

Proof

Proof. Rearranging Definition 0.10:

$$\begin{aligned} \Pr(A \mid B) &= \frac{\Pr(A \cap B)}{\Pr(B)} && \text{(definition of conditional probability)} \\ \Pr(A \cap B) &= \Pr(A \mid B) \cdot \Pr(B) && \text{(multiply both sides by } \Pr(B)) \end{aligned}$$

□

Example 0.13 (Applying the law of conditional probability)**.** Suppose 30% of adults exercise regularly ($\Pr(E) = 0.30$), and among adults who exercise regularly, 60% have low blood pressure ($\Pr(L \mid E) = 0.60$).
Then, by Theorem 0.5, the probability that a randomly selected adult both exercises regularly and has low blood pressure is:

$$\begin{aligned} \Pr(L \cap E) &= \Pr(L \mid E) \cdot \Pr(E) && \text{(law of conditional probability)} \\ &= 0.60 \cdot 0.30 && \text{(substitute the given values)} \\ &= 0.18 && \text{(multiply)} \end{aligned}$$

Theorem 0.6 (Law of total probability)**.** *If $B_1, B_2, \ldots$ is a partition of the sample space, with $\Pr(B_i) > 0$ for every i, then for any event A:*

$$\Pr(A) = \sum_{i=1}^{\infty} \Pr(A \mid B_i) \cdot \Pr(B_i)$$

Proof

Proof. Since $B_1, B_2, \ldots$ partition the sample space, the events $A \cap B_1, A \cap B_2, \ldots$ are mutually exclusive and their union is A. By countable additivity, and then by Theorem 0.5:

$$\begin{aligned} \Pr(A) &= \sum_{i=1}^{\infty} \Pr(A \cap B_i) && \text{(countable additivity for partition of } A) \\ &= \sum_{i=1}^{\infty} \Pr(A \mid B_i) \cdot \Pr(B_i) && \text{(law of conditional probability; } \Pr(B_i) > 0) \end{aligned}$$

□

Theorem 0.7 (Bayes' theorem)**.** *For any two events A and B with $\Pr(A) > 0$ and $\Pr(B) > 0$:*

$$\Pr(A \mid B) = \frac{\Pr(B \mid A) \cdot \Pr(A)}{\Pr(B)}$$

Proof

Proof. By Definition 0.10 and Theorem 0.5:

$$\begin{aligned}
\Pr(A \mid B) &= \frac{\Pr(A \cap B)}{\Pr(B)} && \text{(definition of conditional probability)} \\
&= \frac{\Pr(B \cap A)}{\Pr(B)} && \text{(intersection is commutative: } A \cap B = B \cap A\text{)} \\
&= \frac{\Pr(B \mid A) \cdot \Pr(A)}{\Pr(B)} && \text{(law of conditional probability applied to } \Pr(B \cap A)\text{)}
\end{aligned}$$

□

Example 0.14 (Positive predictive value of a medical test)**.** Suppose a disease test has 99% sensitivity and 99% specificity, and the prevalence of the disease in the population is 7%.
Let D be the event "person has the disease" and $+$ be the event "test is positive". Then:

- $\Pr(+ \mid D) = 0.99$ (sensitivity)
- $\Pr(\neg + \mid \neg D) = 0.99$ (specificity), so the false positive rate is

$$\begin{aligned}
\Pr(+ \mid \neg D) &= 1 - 0.99 \\
&= 0.01
\end{aligned}$$

- $\Pr(D) = 0.07$ (prevalence)

By Theorem 0.7, with the denominator expanded by the law of total probability over the partition $\{D, \neg D\}$:

$$
\begin{aligned}
\Pr(D \mid +) &= \frac{\Pr(+ \mid D) \cdot \Pr(D)}{\Pr(+)} && \text{(Bayes' theorem)} \\
&= \frac{\Pr(+ \mid D) \cdot \Pr(D)}{\Pr(+ \mid D) \cdot \Pr(D) + \Pr(+ \mid \neg D) \cdot \Pr(\neg D)} && \text{(law of total probability)} \\
&= \frac{0.99 \cdot 0.07}{0.99 \cdot 0.07 + 0.01 \cdot 0.93} && \text{(substitute the given values)} \\
&= \frac{0.0693}{0.0693 + 0.0093} && \text{(multiply each term in the numerator and denomina} \\
&= \frac{0.0693}{0.0786} && \text{(add the denominator's two terms)} \\
&\approx 0.88 && \text{(divide)}
\end{aligned}
$$

Even with a highly accurate test (99% sensitive and 99% specific), only about 88% of people who test positive actually have the disease, because the disease prevalence is relatively low (7%).

Exercise 0.1 (Turn a conditional around)**.** Definition 0.10 defines $\Pr(A \mid B)$ in terms of $\Pr(A \cap B)$. Write the same definition with A and B swapped, then eliminate $\Pr(A \cap B)$ between the two.
Your result should express $\Pr(A \mid B)$ using $\Pr(B \mid A)$, $\Pr(A)$ and $\Pr(B)$, and none of them jointly.

Solution

Solution 0.1. Multiplying Definition 0.10 through by its denominator, in each direction,

$$\Pr(A \cap B) = \Pr(A \mid B)\ \Pr(B), \qquad \Pr(A \cap B) = \Pr(B \mid A)\ \Pr(A)$$

The left-hand sides are the same quantity, so the right-hand sides are equal. Dividing by $\Pr(B)$ gives **Bayes' rule** (Theorem 0.7):

$$\Pr(A \mid B) = \frac{\Pr(B \mid A)\ \Pr(A)}{\Pr(B)} \tag{1}$$

Nothing was assumed beyond $\Pr(A) > 0$ and $\Pr(B) > 0$, which the two conditionals need in order to be defined at all.
As Hutchinson emphasizes (Hutchinson 2024), each component carries standard Bayesian terminology:

- $\Pr(A \mid B)$ is the **posterior probability** of A given the observed evidence B.
- $\Pr(B \mid A)$ is the **likelihood** of the evidence B given A.

- $\Pr(A)$ is the **prior probability** of A before seeing B.
- $\Pr(B)$ is the marginal probability of the **evidence**.

From this formula, the direction of effects is transparent: increasing the prior $\Pr(A)$ or the likelihood $\Pr(B \mid A)$ increases the posterior $\Pr(A \mid B)$, while increasing the probability of the evidence $\Pr(B)$ decreases it.

Exercise 0.2 (A test that is right 99% of the time)**.** A screening test for a disease is positive for 99% of people who have it and negative for 99% of people who do not. One person in a thousand has the disease.
A randomly screened person tests positive. What is the probability that they have the disease? Guess first, then compute it with Equation 1.

Solution

Solution 0.2. Write S for having the disease and $+$ for a positive test. The three given numbers are

$$\Pr(+ \mid S) = 0.99, \qquad \Pr(+ \mid \text{not } S) = 0.01, \qquad \Pr(S) = 0.001$$

Equation 1 needs $\Pr(+)$, the overall chance of a positive result. Assemble it with Theorem 0.6, splitting on whether the person is sick:

$$\begin{aligned} \Pr(+) &= \Pr(+ \mid S)\Pr(S) + \Pr(+ \mid \text{not } S)\Pr(\text{not } S) \\ &= (0.99)(0.001) + (0.01)(0.999) \\ &= 0.00099 + 0.00999 \\ &= 0.01098 \end{aligned}$$

Then

$$\begin{aligned} \Pr(S \mid +) &= \frac{\Pr(+ \mid S)\ \Pr(S)}{\Pr(+)} \\ &= \frac{0.00099}{0.01098} \\ &\approx 0.0902 \end{aligned}$$

About **9%**, not the 99% most people guess.
The reason is the **base rate**. Out of 100,000 people screened, about 100 have the disease and 99 of them test positive, while 99,900 do not have it and about 999 of them test positive anyway. The false positives outnumber the true positives ten to one, because there are a thousand times more people available to produce them.
The base rate is fundamental to evaluating screening tests and binary classifiers, and it illustrates why raw accuracy can be misleading on rare events.

The course website's version of this page has an interactive plot here.

> Video lecture
>
> Hutchinson's Probability Refresher (27 min) covers conditional distributions, the law of total probability, the chain rule of probability, and Bayes' rule (Hutchinson, n.d.). The login for the video site is posted on Canvas.

References

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